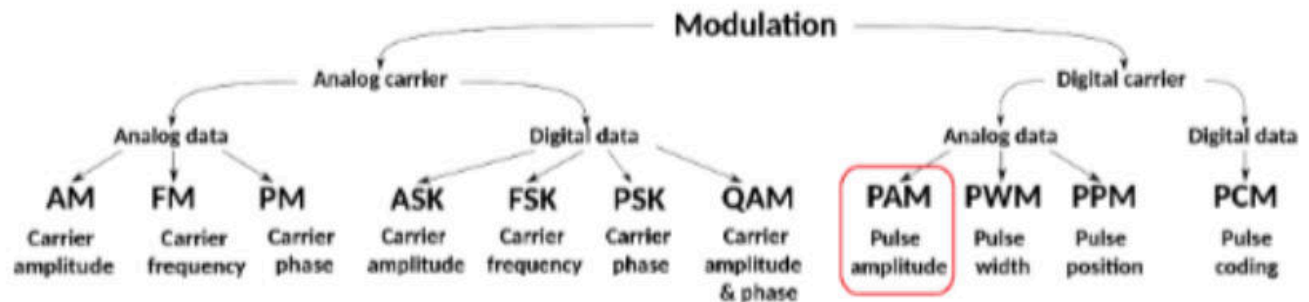


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ETE 3351
Lab 12

DIGITAL SIGNALS PART 2: SAMPLING AND PULSE AMPLITUDE MODULATION (PAM) LAB



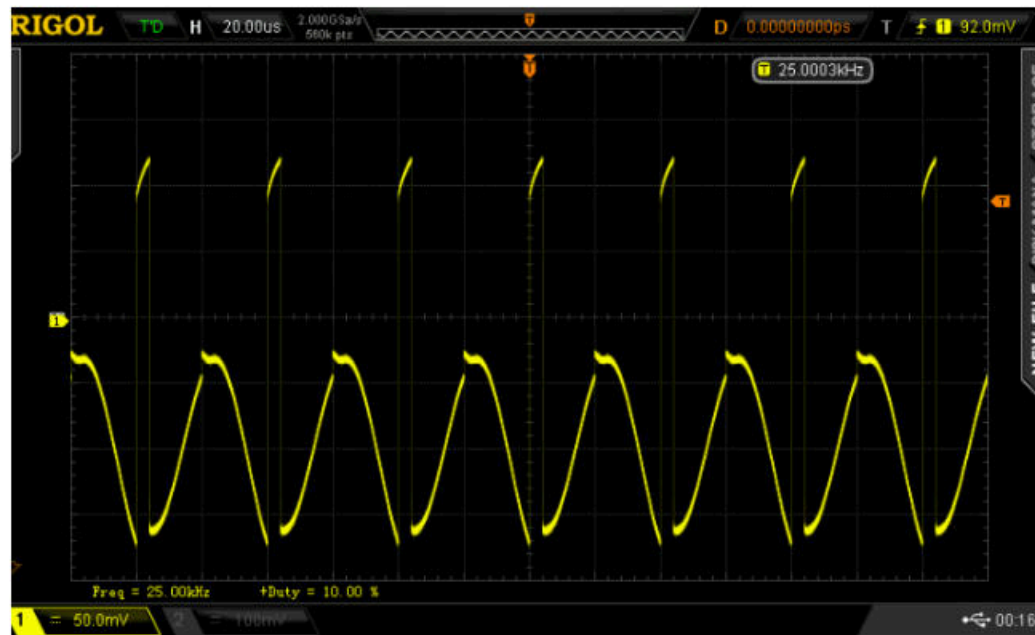
Introduction:

The objective of this lab is to create, analyze and examine a typical sampling signal known as Pulse Amplitude Modulation (PAM) in both the time and frequency domains using a modulation generator. This lab demonstrates how the Nyquist Sampling Theorem plays a critical role in determining how accurately an analog signal can be represented in the frequency domain. By usage of a spectrum analyzer the lab helps illustrate the relationship between sampling frequency, the Nyquist rate and aliasing. Sampling is the process of taking measurements of an analog voltage at regular time intervals and is a key step in analog-to-digital conversion, which was previously discussed in ETE 4201. To accurately capture frequency information a signal must be sampled at least twice its highest frequency, according to the Nyquist theorem. However, higher sampling rates are preferable. In PAM a pulse signal is used to sample the amplitude of the analog signal, creating a sequence of pulses whose heights represent the original signal values.

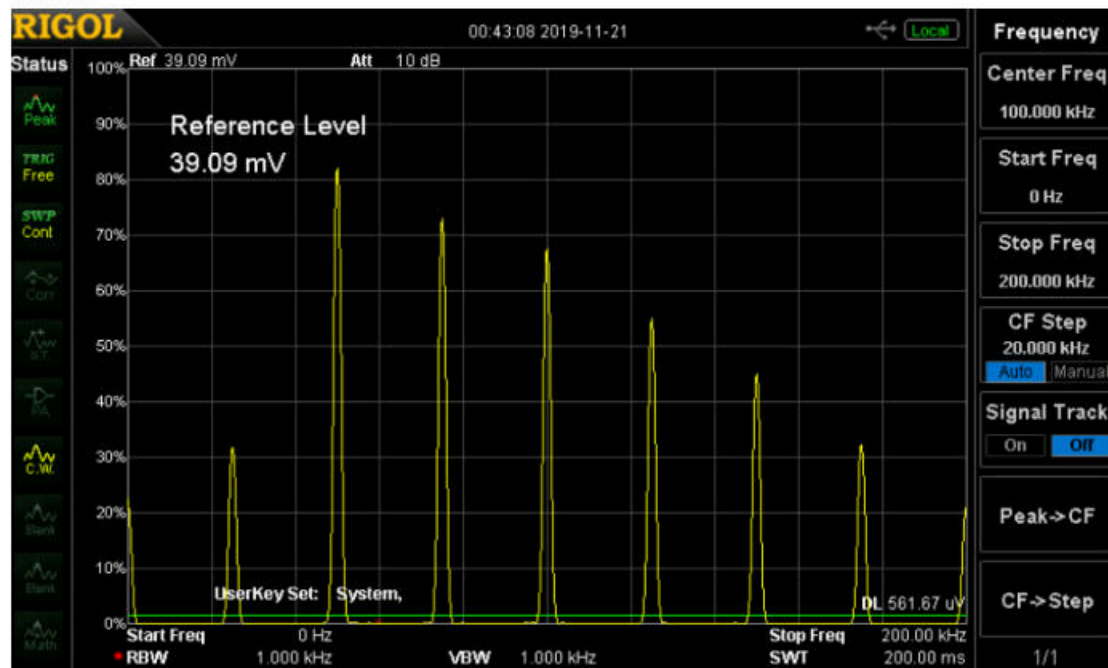
The lab also explores aliasing, which occurs when the sampling frequency is too low and the frequency and/ or shape is misrepresented. Images 1-4 demonstrate just how the sampling rate can affect our data. In the frequency domain, this causes the upper and lower sidebands to overlap making different frequency components appear as a single incorrect signal. This effect is similar to visual aliasing viewable in our daily life, such as when rotating wheels appear to

slow down or move backward when they aren't actually doing so. By varying the sampling frequency the effects of proper sampling, aliasing, and sideband overlap are observed. Additionally similarities between PAM and mixer signals are noted, along with the impact of the sinc ($\sin x / x$) function on the output frequency response.

At nyquist limit:

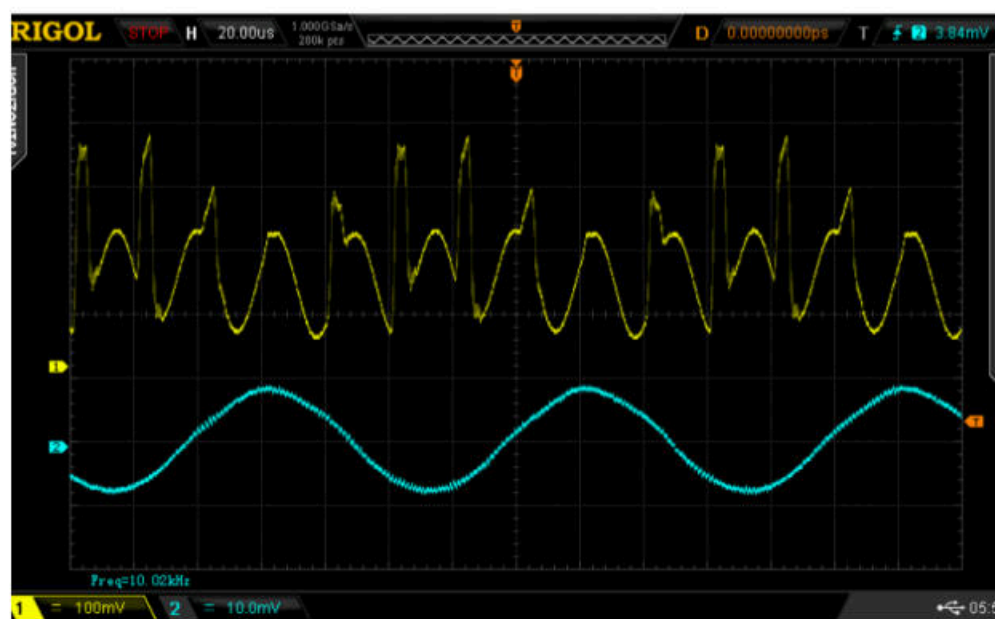


IMG 1.

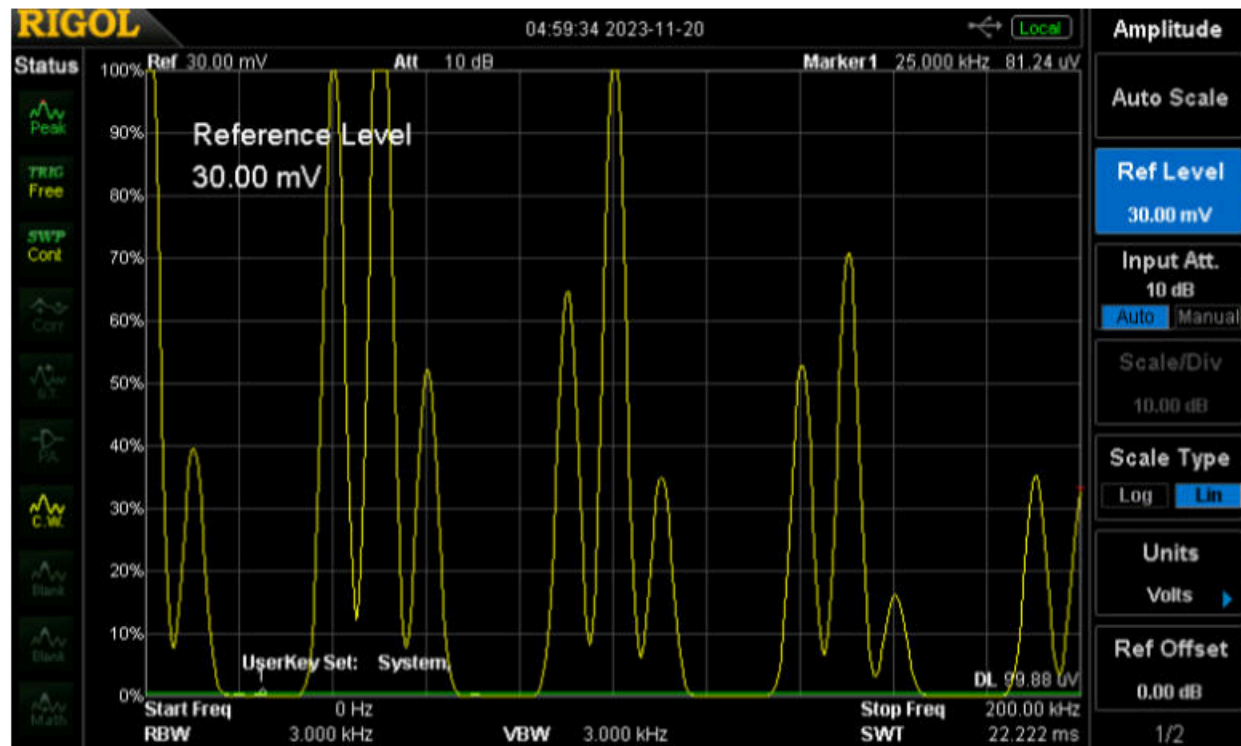


IMG 2.

Beyond the Nyquist limit:



IMG 3.



IMG 4.

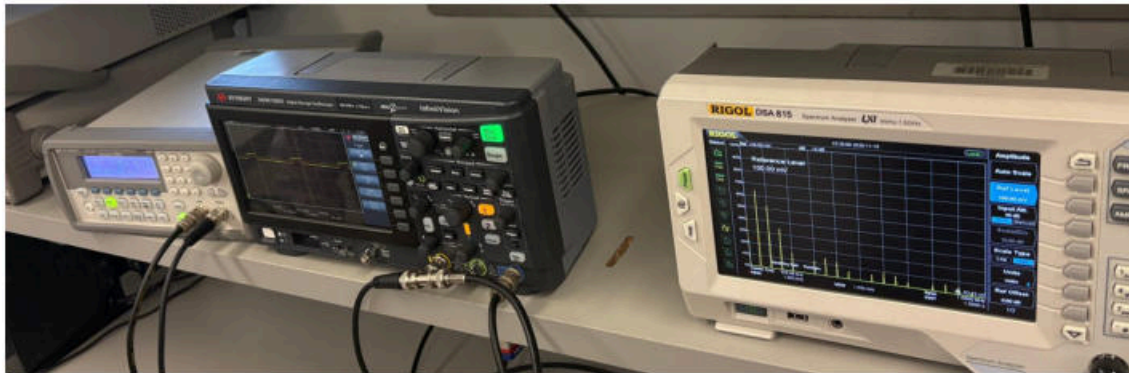
1. We started out with the Rigol ARB Generator connected to the scope and spectrum analyzer.

Spectrum Analyzer (SA) settings:

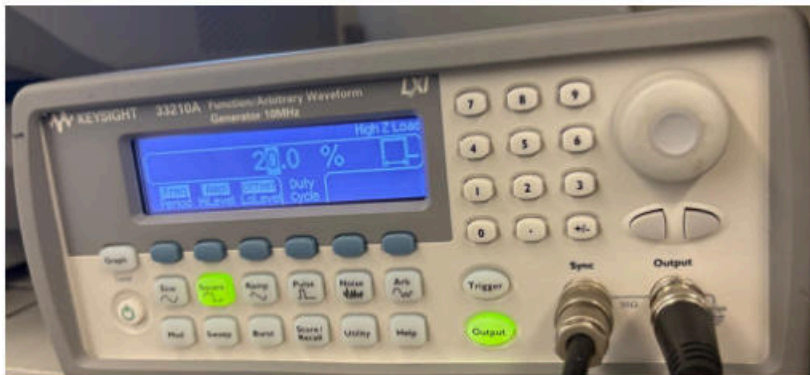
- Start Freq 0Hz
- Stop Frequency 1MHz
- RBW (Resolution Bandwidth) 1kHz or less (remember a lower bandwidth means a slower update but typically a better spectrum display).
- Amplitude Linear (set reference level to 100mV which is the top of the screen),
- input attenuation at 10dB or Auto.

Rigol ARB Generator settings:

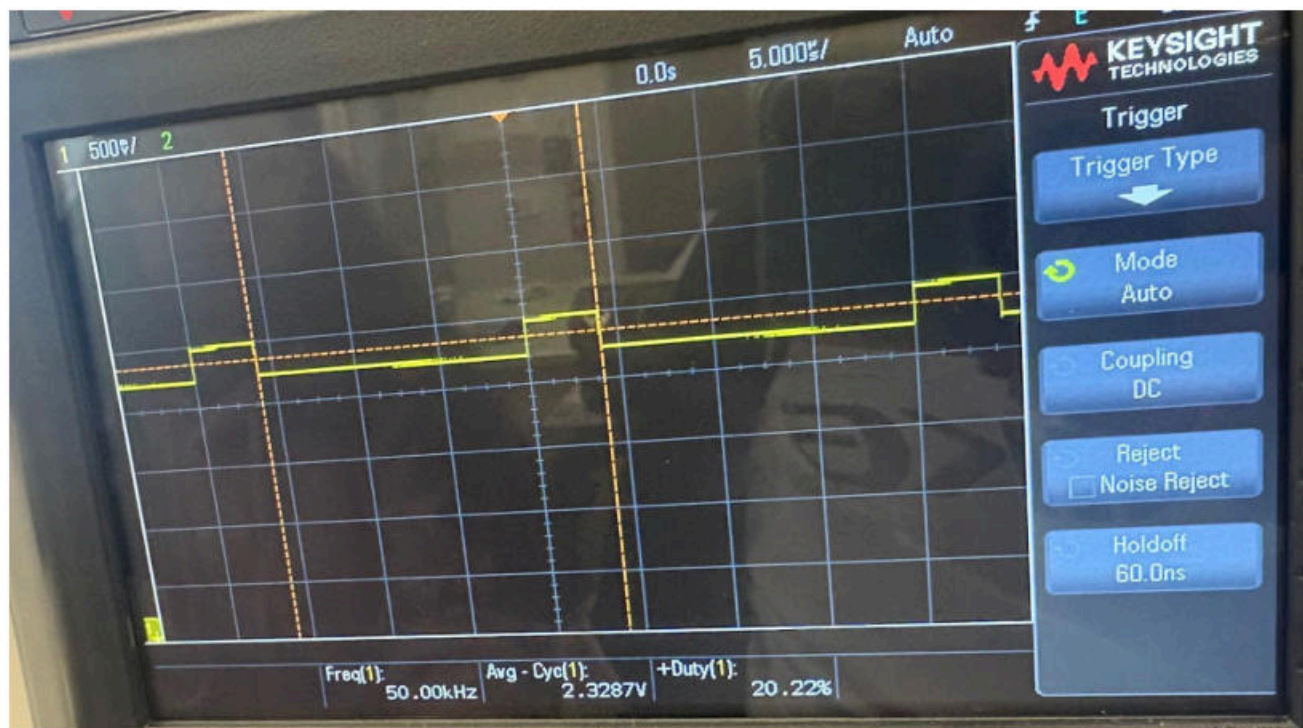
- 50kHz squarewave and a signal of 0.5Vpp,
- Duty cycle at minimum 20%, if the Spectrum Analyzer starts to “beep” please reduce signal level.



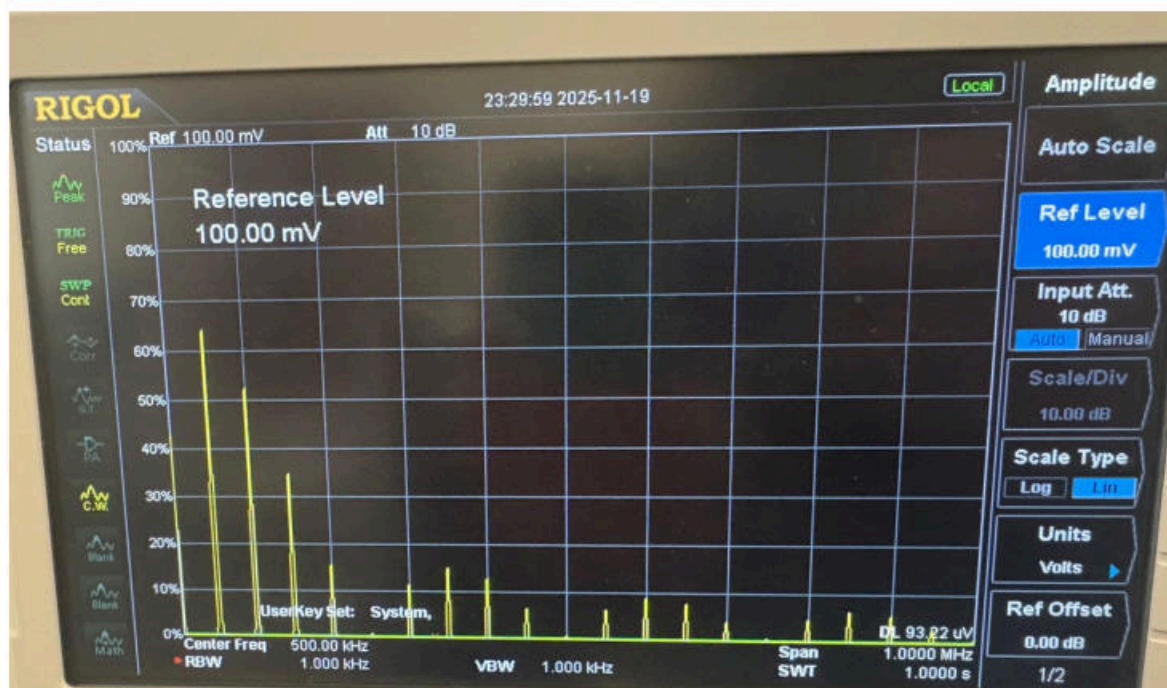
IMG 5.



IMG 6.



IMG 7.

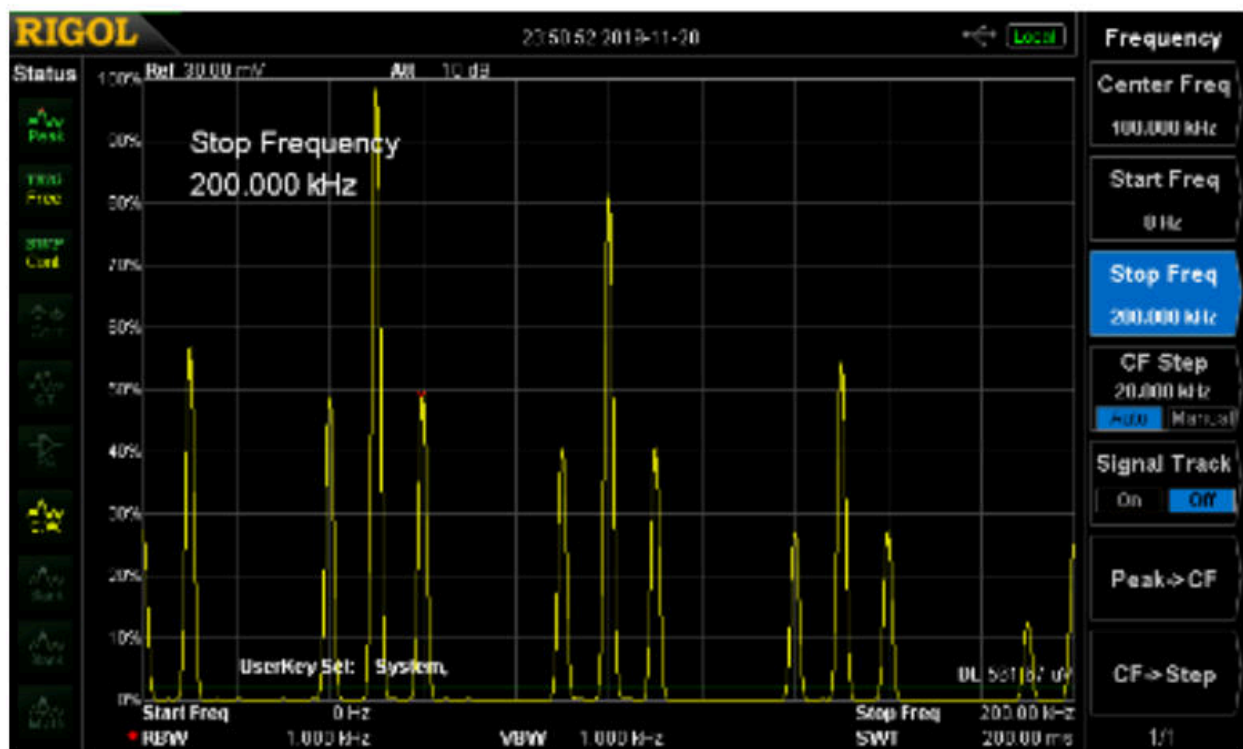


IMG 8.

Initially, the oscilloscope can be triggered in AUTO mode without any issues. However, once modulation is added, obtaining a stable and clear waveform requires using the SYNC output from the ARB generator. The SYNC signal should be connected to Channel 2 or the external trigger input on the oscilloscope, and the scope should be triggered on this signal by adjusting the trigger level using the Trigger Menu and the trigger level adjustment knob. The resulting sampling waveform on the oscilloscope may not appear as a perfect 20% duty-cycle pulse or square wave due to cabling effects and impedance mismatches between the scope and the spectrum analyzer, but this is not critical for the experiment. On the spectrum analyzer, a lower resolution bandwidth (RBW) should be selected to narrow the display bandwidth, which will increase the sweep time but provide a clearer frequency-domain view. The spectrum of the square or pulse waveform will show the characteristic sinc ($\sin x / x$) shape, displayed as an absolute value since the spectrum analyzer measures power. For a 20% duty-cycle signal (one-fifth), the 5th harmonic is very small and essentially zero when viewed in linear mode. Screenshots should be saved for the lab report by using the printer button on the spectrum analyzer to store files to a flash drive, and the Storage menu on the oscilloscope to save images in BMP or JPEG format, with files renamed later on a computer for convenience.

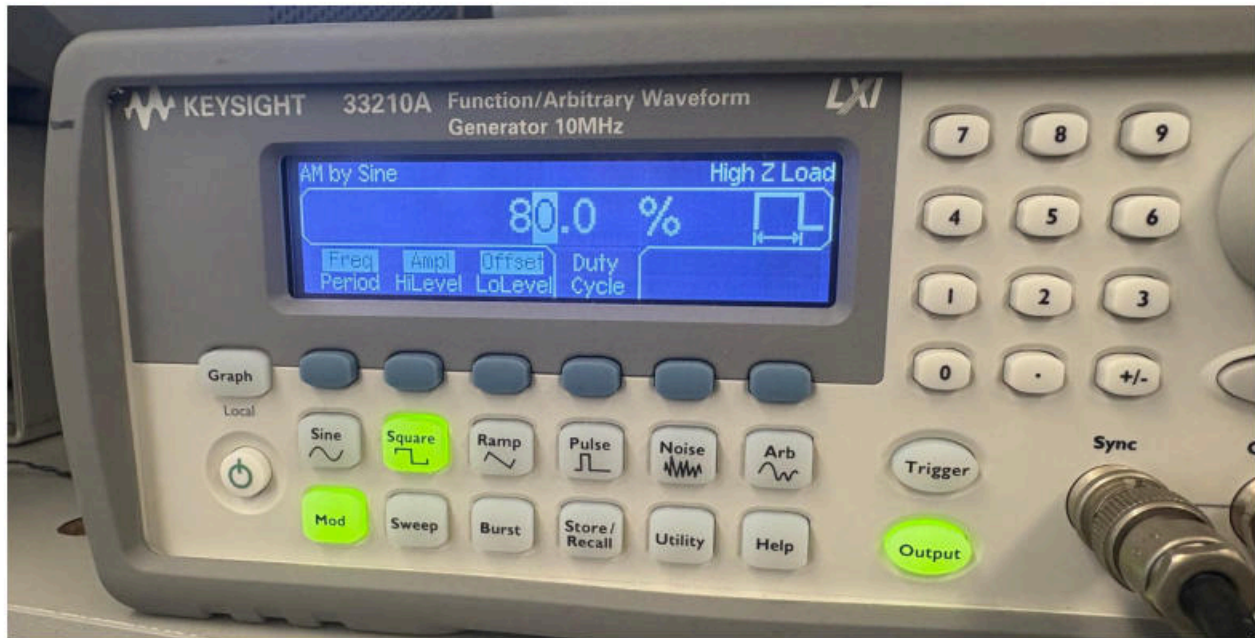
2. Next Turn on the MODulation on the ARB generator: Settings; AM, Sine wave at 10kHz, Modulation percentage 100%.

Spectrum Analyzer view, Reference level adjusted to 30mV, notice for the 50kHz fundamental fs term now have an upper and lower sideband, due to the modulation/multiplication process. With 20% duty cycle the DC component is also represented by the 10kHz signal in the spectrum.

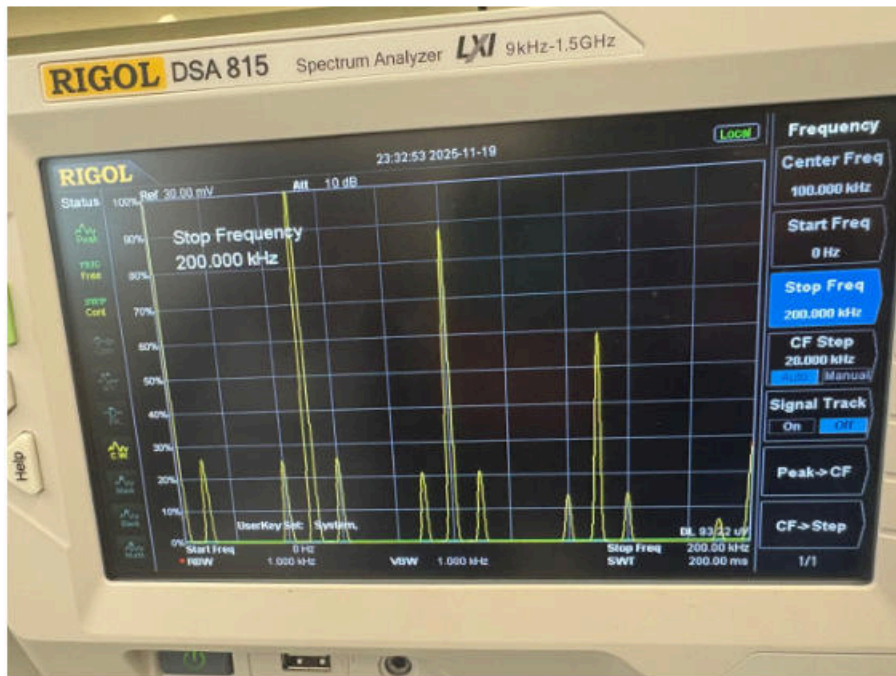


IMG 9. Expected results

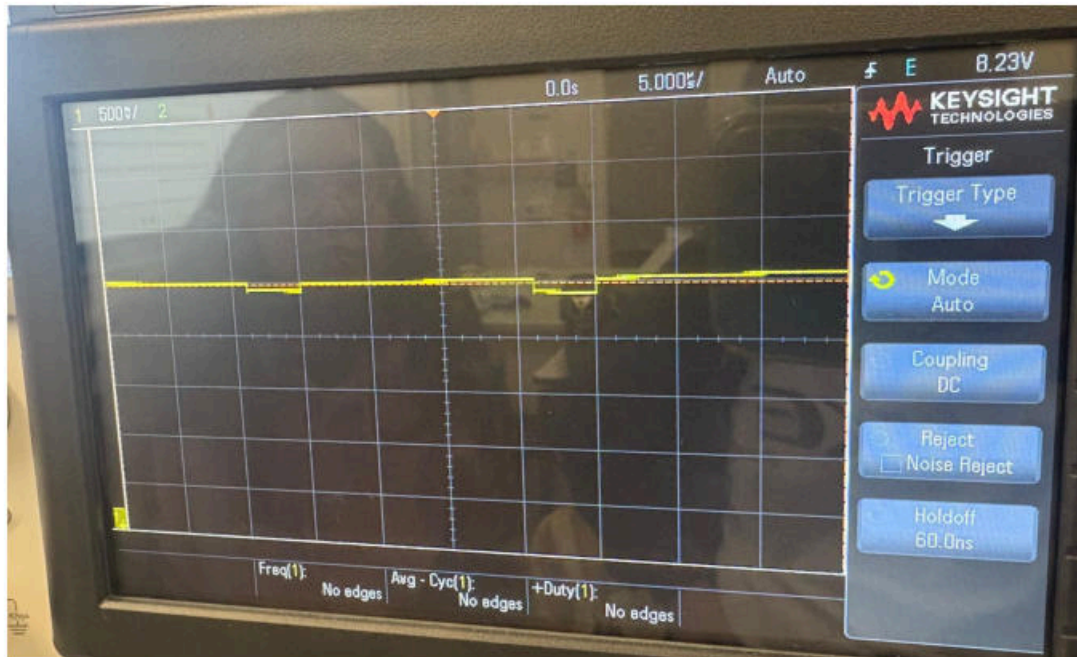
The modulation percentage was capped at 80% due to system limitations, so it was not possible to increase the modulation to 100%. Therefore our results did not exactly match image 9.



IMG 10.

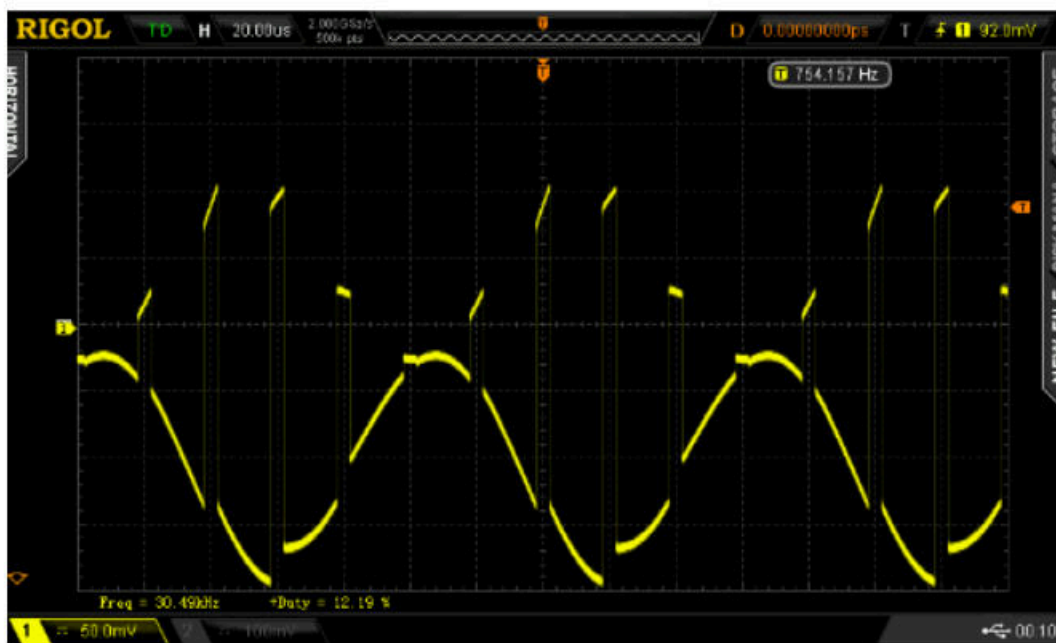


IMG 11. Our results



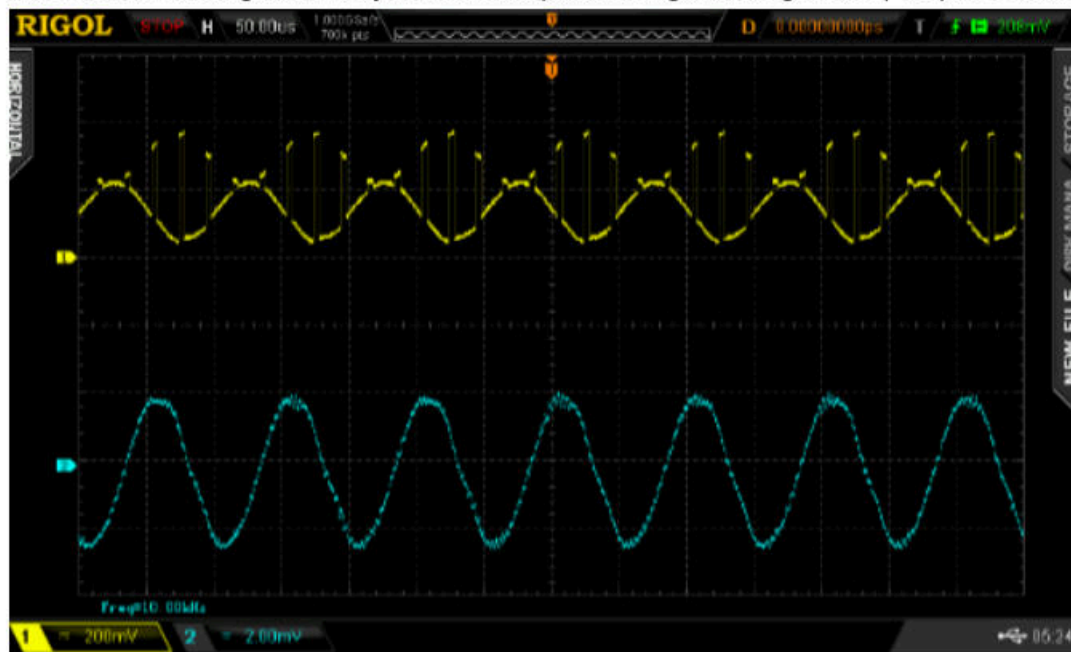
IMG 12. *Our results*

In the time domain view is a typical sampled waveform with a bipolar +1, -1 sampling signal not the typical unipolar 0 and 1 volt signal but this is still a PAM type signal. The sinusoidal amplitude envelope clearly viewable as is its inverse.

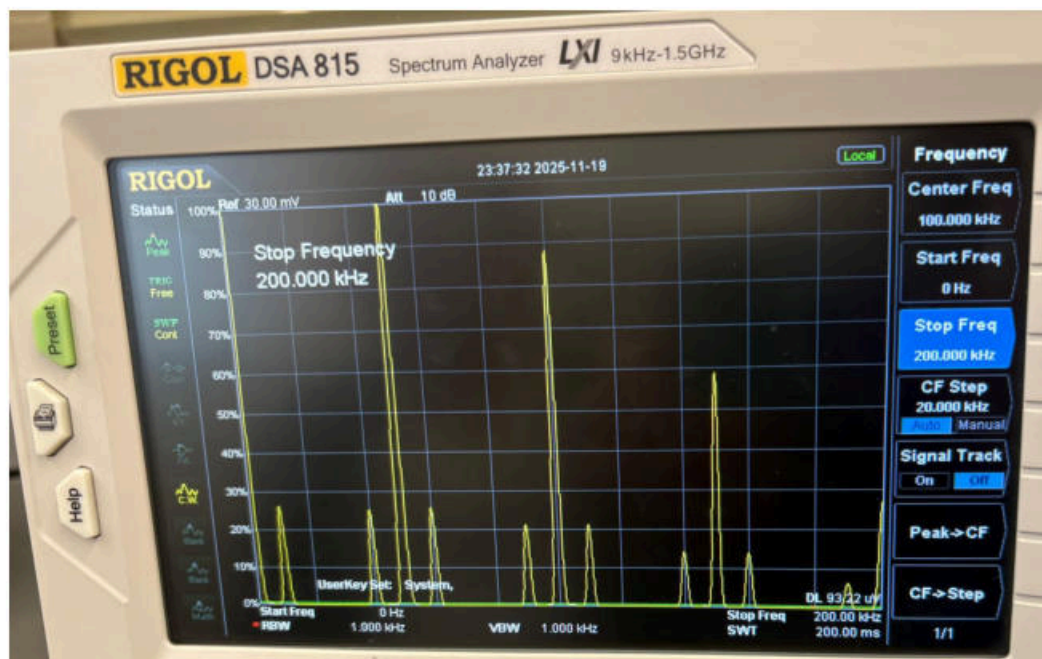


IMG 13. *Image from the laboratory hand out*

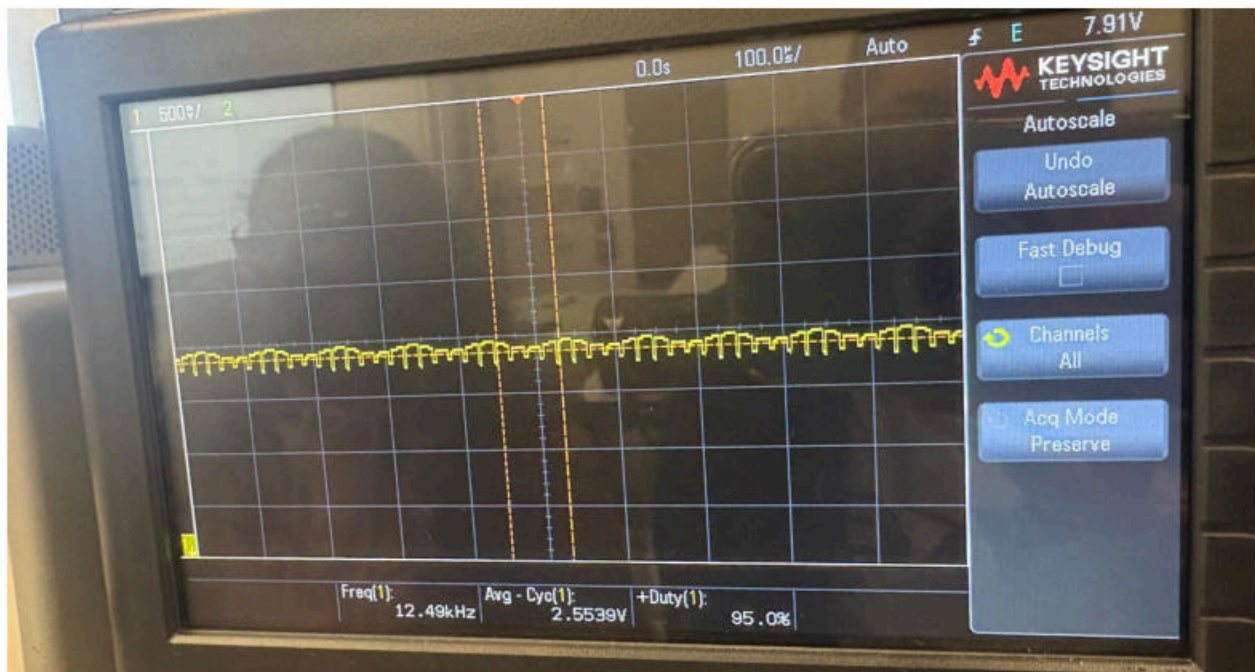
As one might expect looking at the SA frequency display if you put this signal through a low pass filter slightly above 10kHz (12kHz for below) you will see the original 10kHz signal but at these levels the signal is very low and requires a high average rate (>32) to reduce the noise.



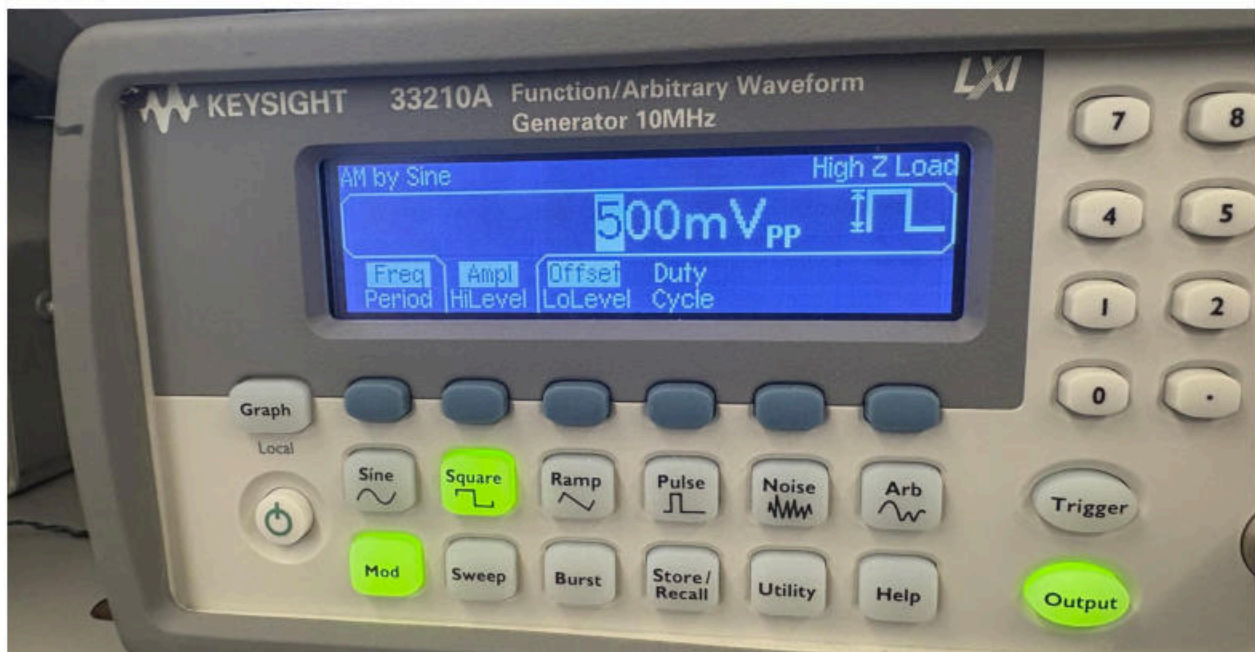
IMG 14. Image from the laboratory hand out
Actually changed to 1vpeak



IMG 15.

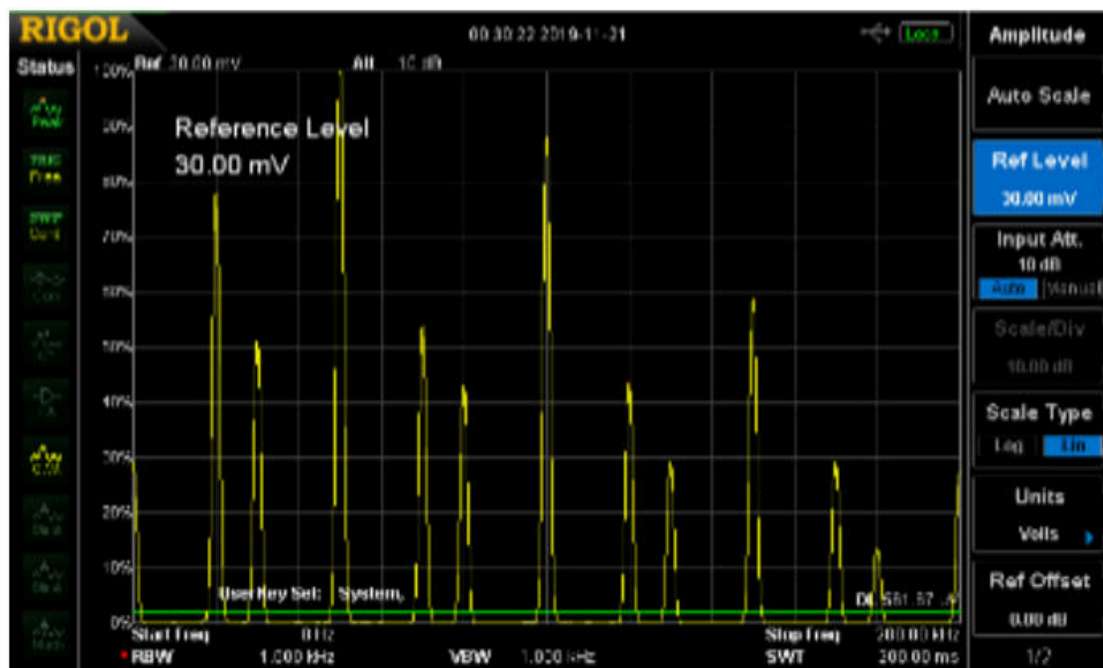


IMG 16.

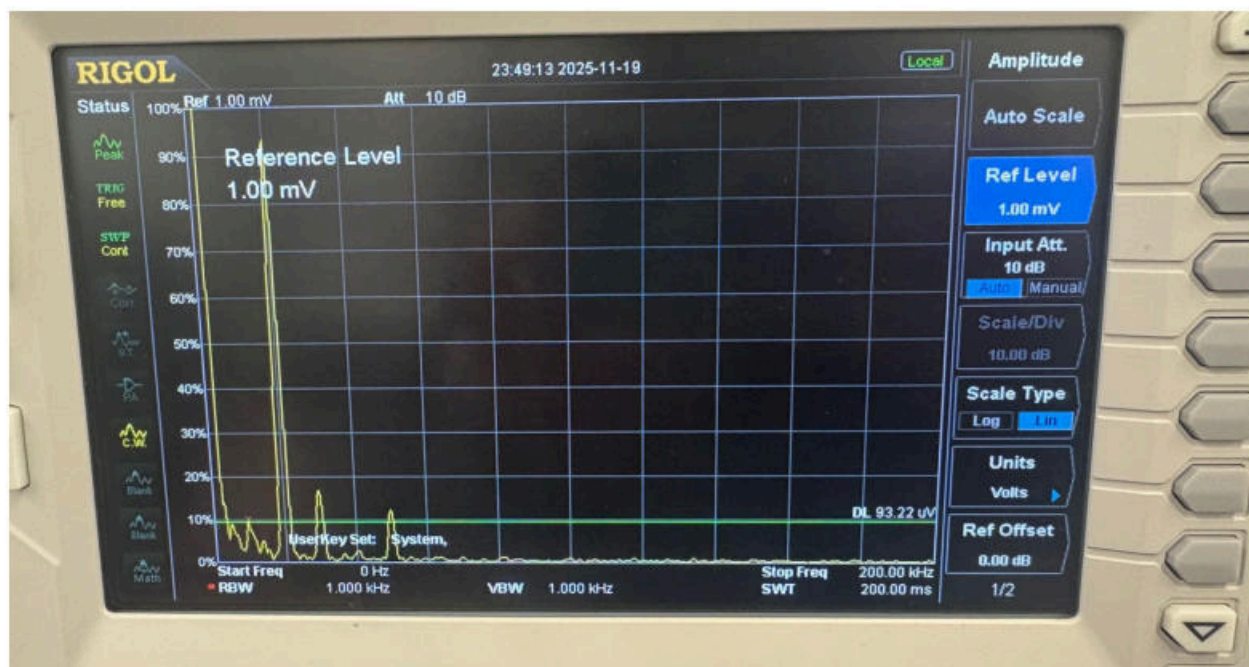


IMG 17.

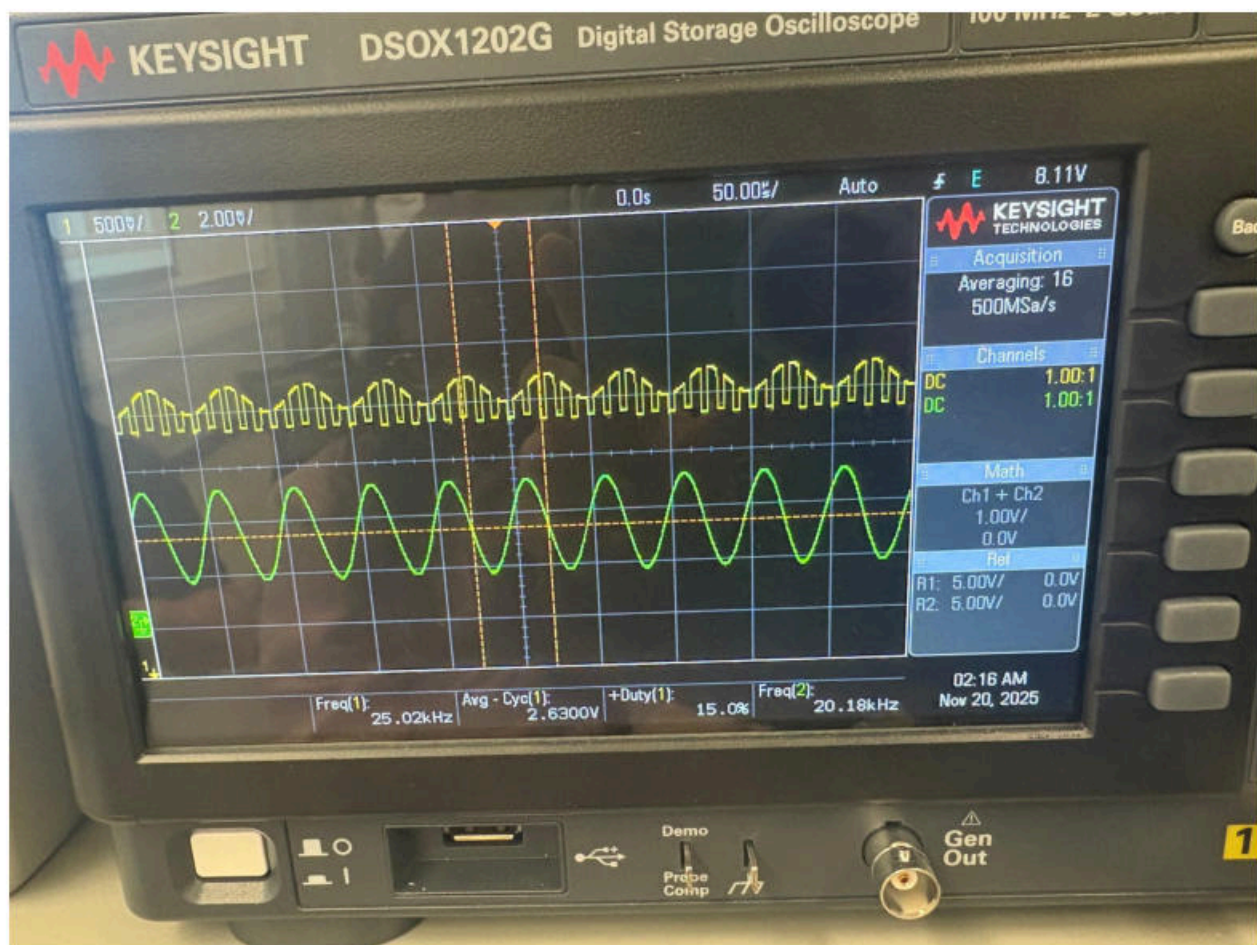
Increasing the modulation frequency to 20 kHz, just below 25 kHz (one-half of the Nyquist sampling frequency of 50 kHz), results in viewable frequency components at 20 kHz and 30 kHz, corresponding to $(50 - 20)$ kHz. Additional components are observed at 50 kHz, 70 kHz $(50 + 20)$ kHz, and 80 kHz $(100 - 20)$ kHz, among others.



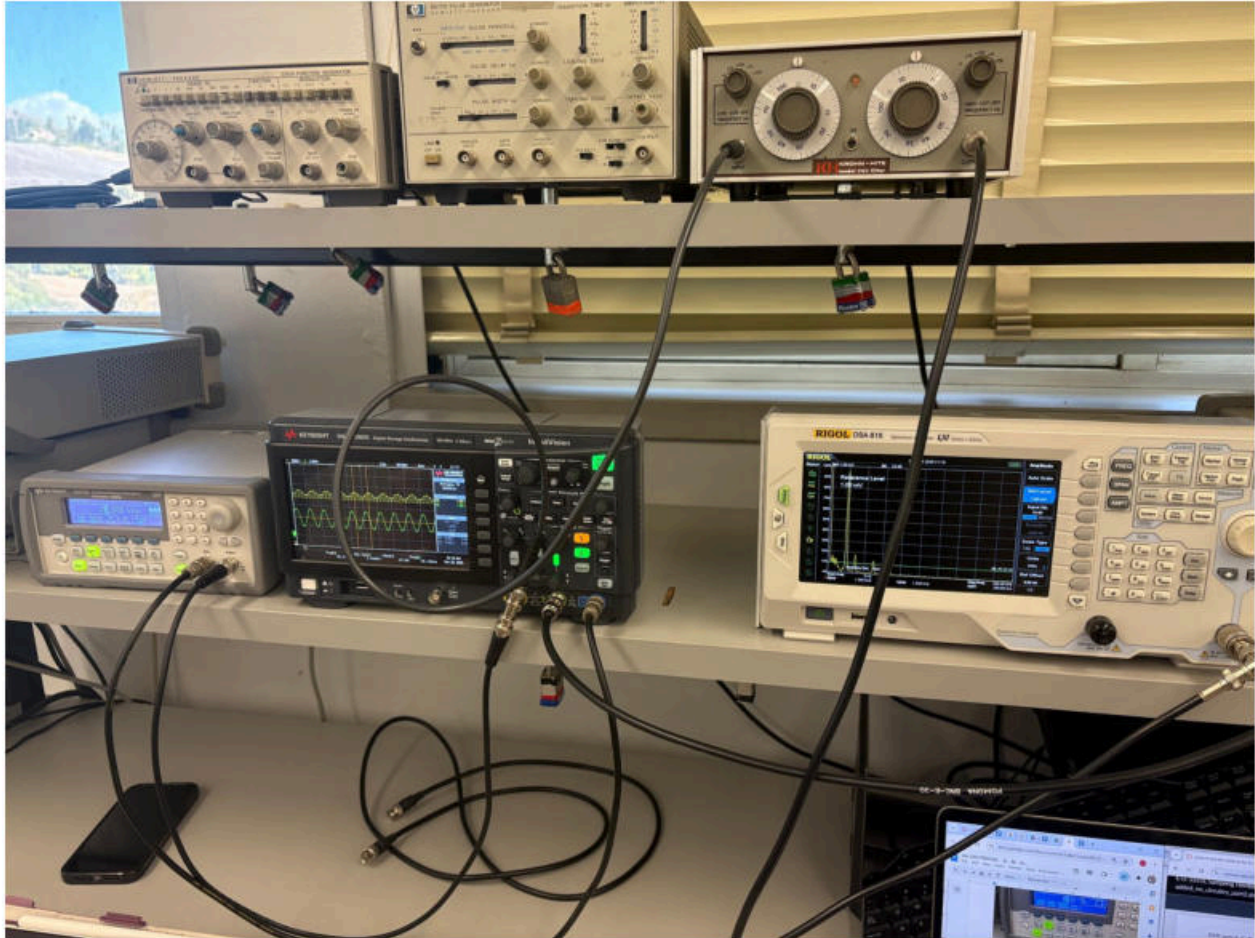
IMG 15. Image from the laboratory hand out



IMG 16



IMG 17



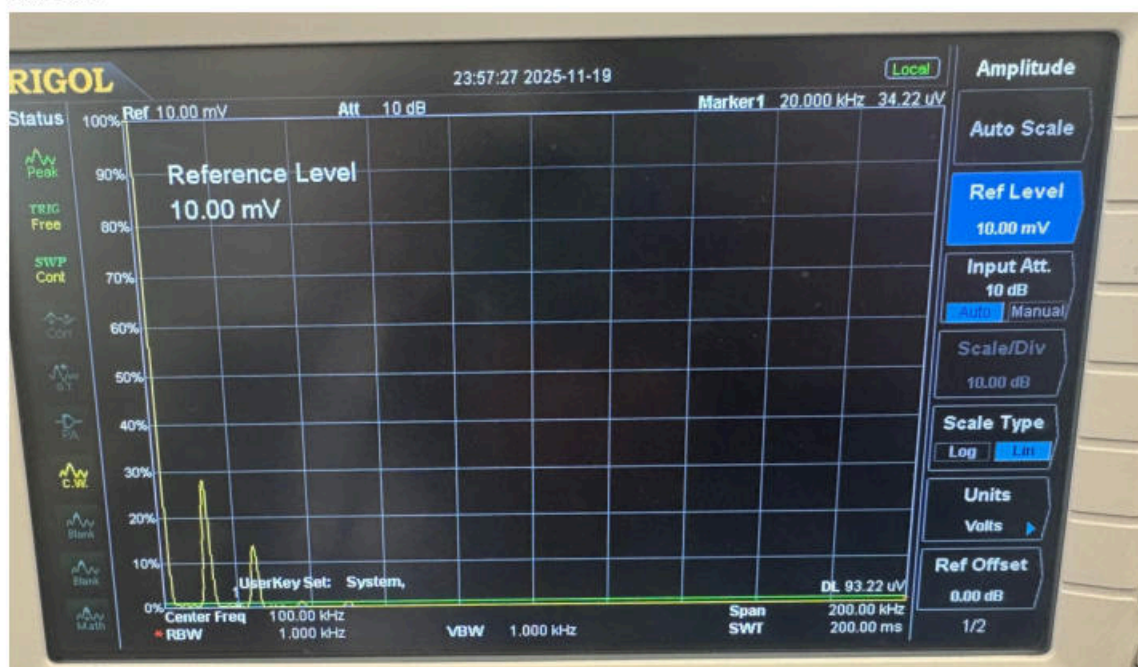
IMG 18

4. At the Nyquist limit when the Modulation frequency is 25kHz exactly $\frac{1}{2}$ the Nyquist sampling frequency of 50KHz, viewable terms are at 25kHz, 50kHz, 75kHz, 100kHz etc. Essentially the 25kHz signal merged to other frequency terms since 25kHz and $(50-25)$ kHz is the same value, as is $(50+25)$ or 75 kHz and $(100-25)$ or 75 kHz, this means we have combined the two signals and have lost information. In the time domain the positive sampled value is essentially constant as the waveform is being sampled at the same time each cycle so the sinusoidal sampled envelope is no longer viewable (constant positive voltage points in figure below).

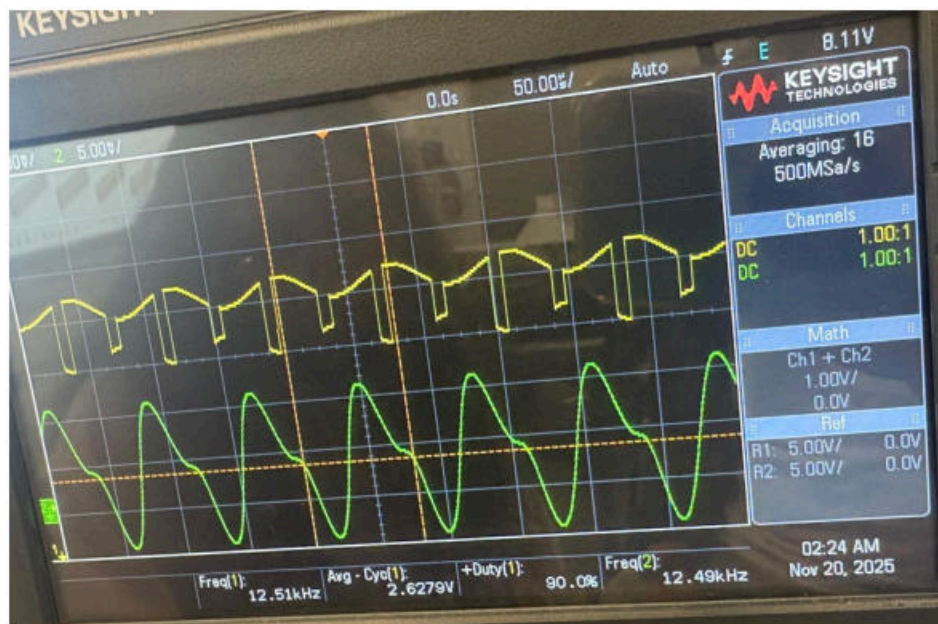
The signal is filtered before sampling because frequency components greater than one-half of the sampling frequency would cause aliasing, highlighting the close relationship between sampling and mixer behavior.



IMG 19



IMG 20

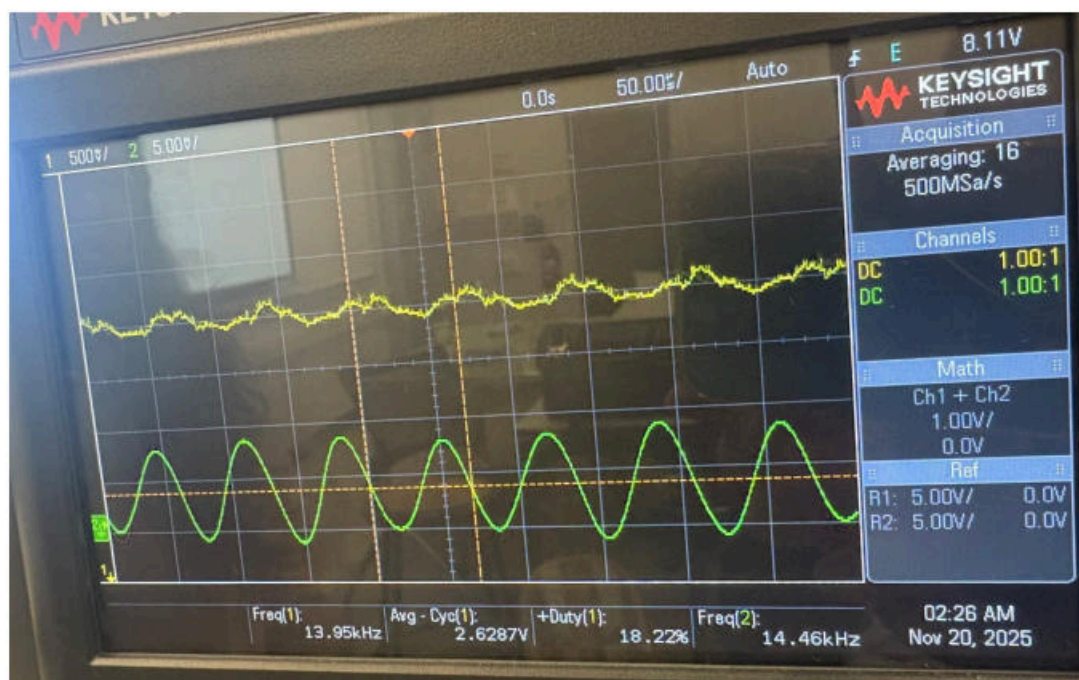


IMG 21

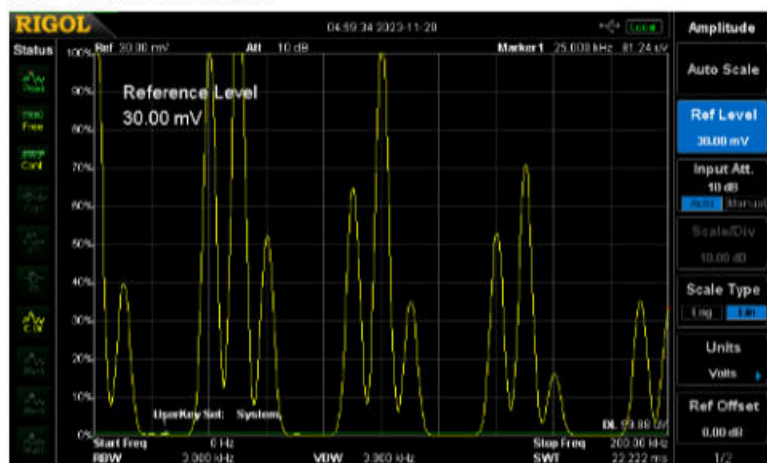
5. Beyond the Nyquist limit, increasing the modulation frequency to 40 kHz, which is 15 kHz above the Nyquist frequency of 25 kHz, causes the sidebands to reappear in a different order with respect to the original sampling frequency components. In the picture below, the 40 kHz term appears at the 10 kHz location (50 kHz – 40 kHz) when filtered and averaged, meaning the original 40 kHz frequency is now observed at 10 kHz. This frequency component is referred to as a frequency alias, and this condition should be avoided by keeping input signals below the Nyquist frequency of $f_s/2$. The full time-domain view can be confusing due to the presence of many combined sampled or mixed signals.



IMG 22. System set up



IMG 25 time domain



IMG 26. Expected results

Conclusion

In this lab, sampling and Pulse Amplitude Modulation (PAM) were used to study how an analog signal is represented in both the time and frequency domains. By changing the sampling frequency it was observed that sampling above the Nyquist rate allows the original signal to be viewed correctly, while sampling below the Nyquist rate causes aliasing. The spectrum analyzer showed how sampled signals create repeated frequency components and how sidebands can overlap when the sampling frequency is too low. In both AM and PAM, information is carried in amplitude variations. In AM, the amplitude of the carrier changes in proportion to the message signal. In PAM, the amplitude of each pulse changes in proportion to the message signal. We see that PAM has upper and lower sidebands when looking in the frequency domain. PAM also indirectly relates to FM because sampling creates repeated frequency components and aliasing effects similar to mixing and frequency shifting seen in FM systems. This lab also demonstrated the importance of proper sampling frequency selection and filtering to prevent aliasing and accurately represent signals in digital systems.